Experiment 9

Experiment to understand Matplotlib library use cases in Data Science through visualization

Aim:

To understand Matplotlib library use cases in Data Science through visualization

Code:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.read_csv('Iris_Dataset.csv')

print(data.info())
print(data.describe())
print(data.value_counts('variety'))

sns.countplot(x='variety', data=data)
plt.show()

dummies = pd.get_dummies(data.variety)
FinalDataset = pd.concat([pd.get_dummies(data.variety), data.iloc[:, [0, 1, 2, 3]]], axis=1)
print(FinalDataset.head())

sns.scatterplot(x='SepalLengthCm', y='SepalWidthCm', hue='variety', data=data)
plt.show()

sns.scatterplot(x='PetalLengthCm', y='PetalWidthCm', hue='variety', data=data)
plt.show()

sns.pairplot(data, hue='variety', height=3)
plt.show()

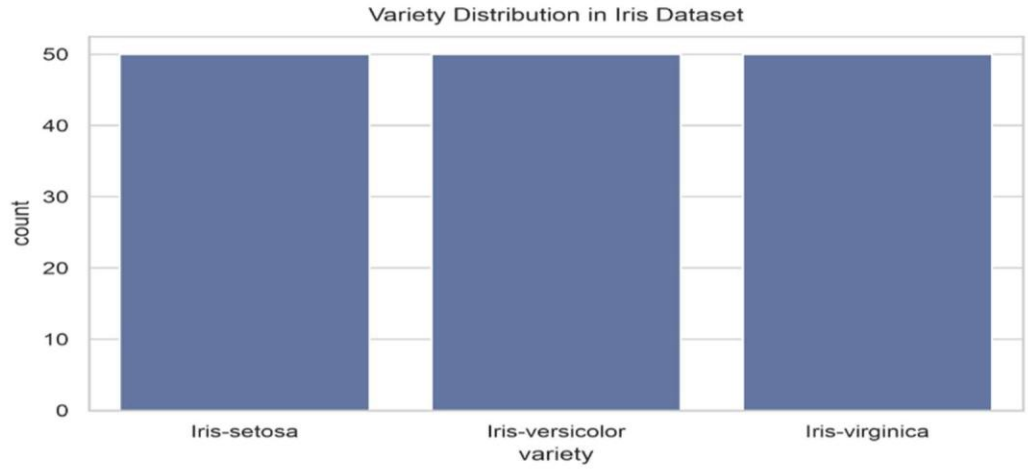
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'PetalLengthCm').add_legend()
plt.show()

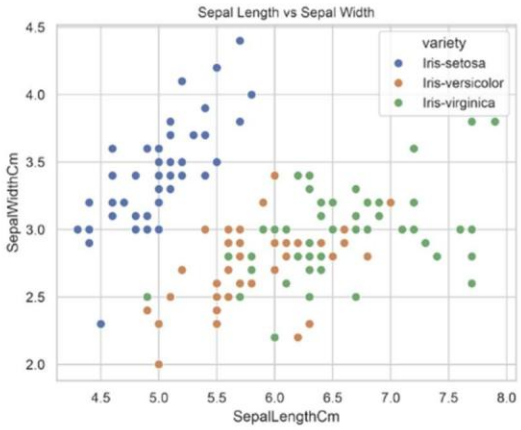
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'PetalWidthCm').add_legend()
plt.show()

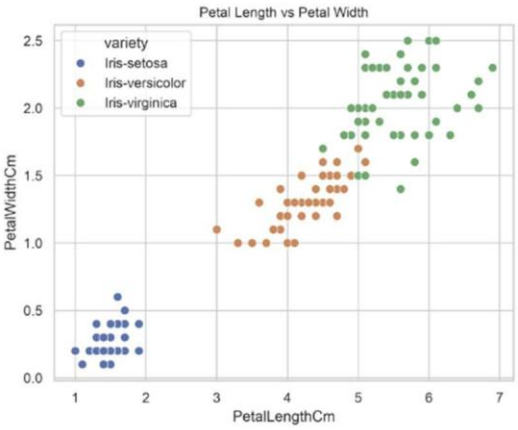
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'SepalLengthCm').add_legend()
plt.show()

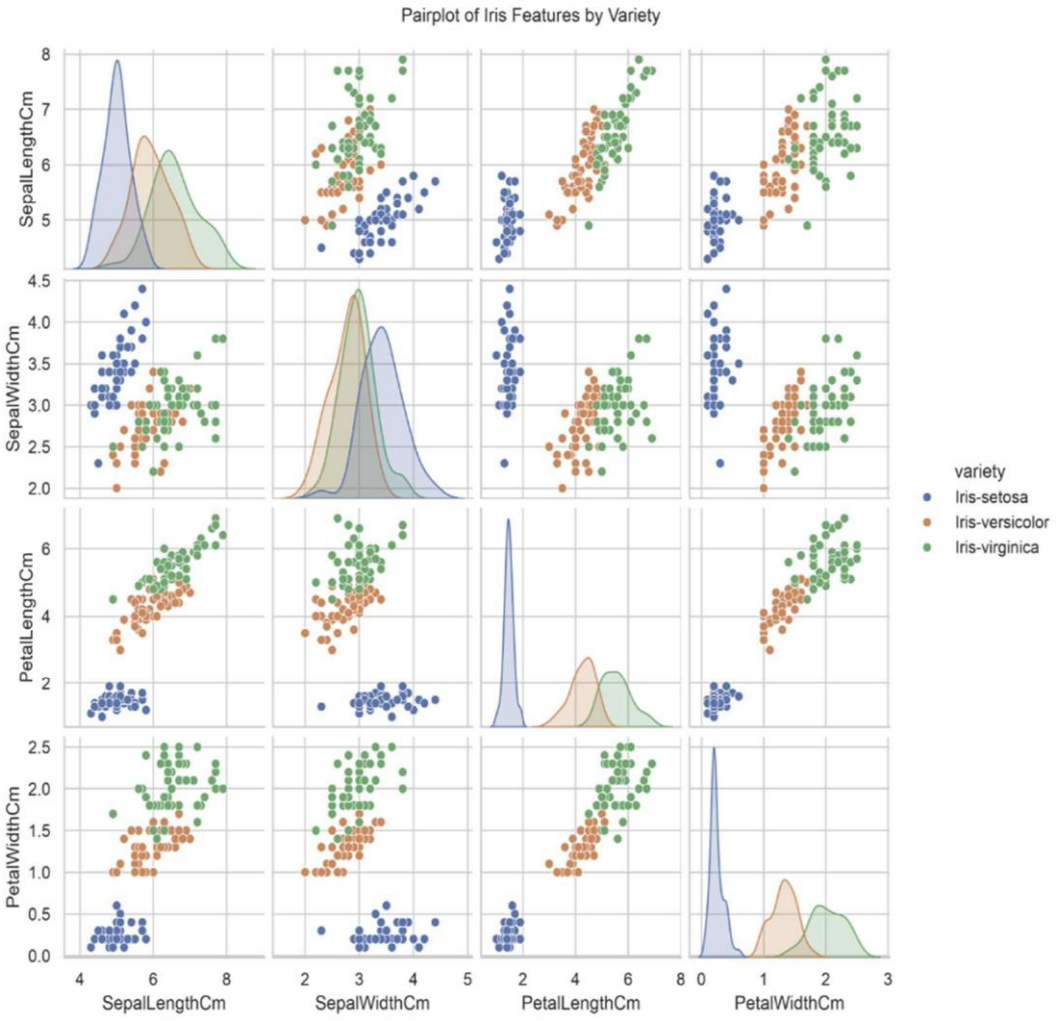
sns.FacetGrid(data, hue='variety', height=5).map(sns.histplot, 'SepalWidthCm').add_legend()
plt.show()
```

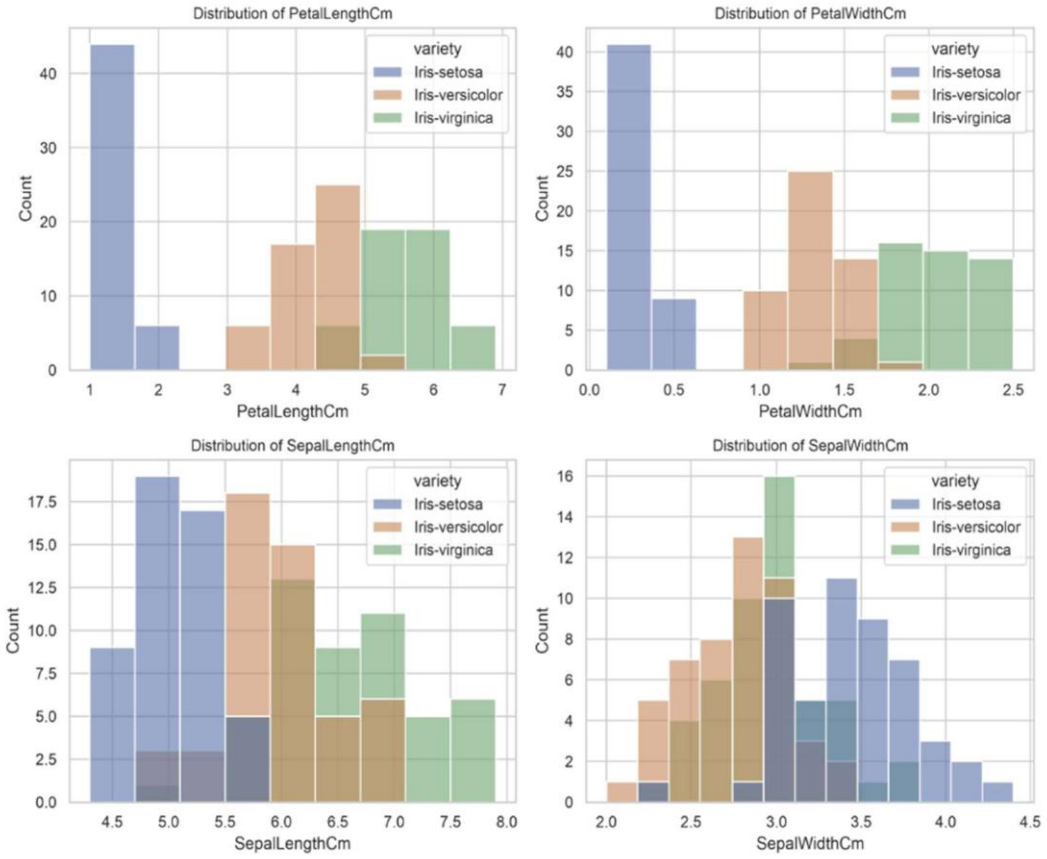
Output:


Variety Distribution in Iris Dataset
count
50
40
30
20
10
0
Iris-setosa
Iris-versicolor
Iris-virginica
variety


Sepal Length vs Sepal Width
variety
Iris-setosa
Iris-versicolor
Iris-virginica
SepalWidthCm
SepalLengthCm
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0


Petal Length vs Petal Width
variety
Iris-setosa
Iris-versicolor
Iris-virginica
PetalWidthCm
PetalLengthCm
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
4
5
6
7


Pairplot of Iris Features by Variety
SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
variety
Iris-setosa
Iris-versicolor
Iris-virginica



Result:

Matplotlib and Seaborn visualizations on the Iris dataset successfully revealed distinct class separability. Pairplots showed that Setosa is linearly separable from Versicolor and Virginica, while heatmaps confirmed strong positive correlation between petal length and petal width.